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# -*- coding: utf-8 -*-
"""
Created on Sun Sep  2 17:23:58 2018

@author: jfowl


$$x''(t) + 2x'(t) + x = 5e^{-2t} + t$$

with initial conditions:
    
$$x(0) = 2$$

    
$$x'(0) = 1$$


with analytic solution:
    
$$x = -e^{-t} + 9te^{-t} + 5e^{-2t} + t - 2$$

"""

import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import odeint

nt = 1000
tstart = 0
tend = 20

tvec = np.linspace(tstart, tend, nt)

def analytic_sol(t):
    first = -np.exp(-t)
    second = 9*t*np.exp(-t)
    third = 5*np.exp(-2*t)
    fourth = t
    fifth = -2
    x = first + second + third + fourth + fifth
    return x

def ode_model(X, t):
    xdot1 = X[1]
    xdot2 = -2*X[1] - X[0] + 5*np.exp(-2*t) + t
    return [xdot1, xdot2]

x0 = [2, 1]
x = odeint(ode_model, x0, tvec)

fig = plt.figure()
ax = fig.add_subplot(111)
ax.plot(tvec, [analytic_sol(t) for t in tvec], 'r', linewidth=5,
        label='Analytic Solution')
ax.plot(tvec, x[:, 0], '-.k', linewidth=2, label='Numerical Solution')
plt.xlim([min(tvec), max(tvec)])
plt.ylim([min(x[:, 0]), max(x[:, 0])])
handles, labels = ax.get_legend_handles_labels()
lgnd = ax.legend(handles, labels, loc='upper left',
                 bbox_to_anchor=(0, 1))
plt.xlabel('t-Value', fontsize=16)

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plt.ylabel('x-Value', fontsize=16)
plt.title('Plot of  $x''(t) + 2x'(t) + x = 5e^{-2t} + t$ ', fontsize=18)
plt.grid(b=True, which='both')
fig.savefig('ME426_HW3_21b', bbox_extra_artists=(lgnd,), bbox_inches='tight')
plt.show()
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